

INTERNSHIP REPORT

INTERNET OF THINGS

Submitted By

LOGESWARAN P	23DR27
PRAGADEESHWARAN S	23DR34
SHARVANTHVADIVELAN	23DR45
SHARVESH S	23DR46

Under the guidance of

Ms.P.ABIRAMI

In partial fulfillment of the requirement for the award of

DIPLOMA IN ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



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DEPARTMENT OF COMPUTER NETWORKING

PSG POLYTECHNIC COLLEGE

(Autonomous and an ISO 9001:2018 certified Institution)

COIMBATORE – 641 004

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CERTIFICATE OF COMPLETION

This is to certify the below student on completing the internship on the topic

INTERNET OF THINGS

LOGESWARAN P	23DR27
PRAGADEESHWARAN S	23DR34
SHARVANTHVADIVELAN	23DR45
SHARVESH S	23DR46

This is to certify that the Internship Report entitled

RFID ATTENDANCE SYSTEM

has been submitted by

SHARVANTHVADIVELAN - 23DR45

In partial fulfillment of the award of

DIPLOMA IN ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Of the State Board of Technical Education,

Government of Tamil Nadu,

During the academic year of 2024 - 2025

Ms.P.Abirami,

Faculty Guide

Dr. S. Brindha,

Head of the Department

Certified that the candidate was examined by us in the completion of the Internship Viva Voce Examination held on

Internal Examiner

External Examiner

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ABSTRACT

The RFID Attendance System project presents an innovative solution to streamline and automate attendance tracking by integrating embedded systems, RFID technology, and microcontroller-based processing. Traditional attendance methods are prone to manual errors, time consumption, and the potential for proxy entries, leading to inefficiencies in academic and organizational environments. This project addresses these challenges by employing a contactless, real-time attendance system using RFID readers and smart identification tags.

The system utilizes an RFID module (RC522) to scan individual RFID cards, each uniquely associated with a student or employee. This data is processed by an Arduino UNO microcontroller, which verifies the identity and records the attendance. Feedback mechanisms such as an LCD display and a buzzer enhance the user experience by providing immediate visual and audio confirmation upon successful scans. For unmatched or unauthorized entries, appropriate alerts are displayed.

Designed with modularity in mind, the system supports future enhancements such as real-time clock (RTC) integration for time-stamping, SD card logging, or even cloud connectivity using ESP32 for remote monitoring. The use of low-power, open-source hardware components ensures that the system remains both cost-effective and scalable.

By combining RFID technology with embedded control and optional wireless integration, the RFID Attendance System enhances efficiency, reduces manual workload, and ensures accurate record-keeping. It is ideal for schools, colleges, and offices aiming to modernize their administrative processes. This project exemplifies how embedded electronics and IoT can be applied to solve real-world problems in an affordable and practical manner.

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CHAPTER 1

INTRODUCTION

1.1 ABOUT THE COMPANY

Enthu Technology Solutions India Pvt Ltd, based in Coimbatore, Tamil Nadu, is a leading provider of embedded systems and IoT-based product solutions. Founded in 2007 and incorporated as a private limited company in 2015, the firm focuses on end-to-end product design, development, and manufacturing, serving a wide range of industrial and academic sectors.

The company provides an end-to-end suite of services including hardware design, embedded system development, mobile application creation, cloud services, and complete manufacturing support. They serve startups, SMEs, educational institutions, and large enterprises across domains like smart cities, agriculture, industrial automation, automotive, and smart buildings. Enthu Tech is known for its robust design capabilities, reliable product development, and adherence to international quality standards. The company is ISO 9001:2015 and ISO 14001 certified and is a member of the LoRa Alliance, reflecting its commitment to high standards and futureforward technologies



Fig 1.1 : Enthu Technology Solutions Company

1.2 Certifications & Memberships:

- ISO 9001:2015 (Quality Management)
- ISO 14001:2015 (Environmental Management)
- IATF 16949:2016 (Automotive Quality)

- Member of the **LoRa Alliance®** (since 2021)

1.3 Innovations of ENTHU TECH

Enthu Tech's innovations are at the forefront of the embedded and IOT space.

These include:

1. thingZmate Platform

A cloud-enabled IoT platform that enables device management, data visualization, and predictive analytics. It is widely used in smart water management, cold-chain monitoring, and precision agriculture.

2. PCBMATE® and SMTMATE®

These are in-house developed PCB prototyping and SMT assembly services tailored for product developers and startups. They have streamlined the prototyping process, enabling quicker go-to-market times.

3. akvodrops™

A smart water meter that allows remote monitoring of water usage, promoting conservation and efficient utility management. It has been deployed in campuses and residential societies.

4. EMS (Electronics Manufacturing Services)

Enthu Tech provides turnkey manufacturing services including PCB design, fabrication, enclosure design, component assembly, testing, and product packaging.

5. Smart IoT Solutions

Applications include smart farming solutions, cold storage monitoring, energy-efficient lighting systems, air quality monitoring, and remote diagnostics

1.4 History of ENTHU TECH

2007 - Formation as a startup focused on student training in embedded systems and robotics.

2009 - Expanded services to include workshops and industrial training for engineering colleges across South India.

2012 - Entered into the embedded design space with in-house prototyping for custom requirements.

2015 - Incorporated as a Private Limited company. Diversified into IOT and EMS, launching PCBMATE® and SMTMATE®.

2016 - Began full-stack IOT development, integrating cloud, firmware, mobile, and analytics.

2019 - Filed patents for internally developed smart IOT solutions.

2020 - Launched thingZmate® platform for integrated AIOT deployments.

2021 - Became a member of the LoRa Alliance; developed solutions compatible with LoRaWAN and NB-IOT.

2022 - Participated in GITEX Global, Dubai, showcasing India-made IOT platforms.

2023 - Signed MoU with Israeli startup GorillaLink to build satellite-based IOT systems

1.5 Core Services:

- Embedded system and IoT product development
- PCB prototyping, SMT assembly, and enclosure design
- Cloud integration, mobile and web application development

- Product testing, validation, and deployment

1.6 Manufacturing Capabilities:

Enthu Tech houses advanced in-house infrastructure, including SMT lines, PCB assembly setups, and enclosure manufacturing units, making it a full-service Electronic Manufacturing Services (EMS) provider.

1.7 IOT Specialization:

The company is recognized for its expertise in:

- LoRaWAN®, NB-IOT, and satellite IOT technologies
- Partnered with GorillaLink to develop satellite-based IOT solutions
- Served both government and private sectors in India and internationally

1.8 Academic & Training Initiatives:

Under its Enthu Academic Solutions (EAS) division, the company:

- Establishes IoT/embedded labs in colleges
- Offers hands-on training, internships, faculty development programs
- Partners in AICTE-IDEA Lab initiatives and R&D projects

1.9 Industry Presence & Events:

Participated in GITEX Global Dubai 2022, showcasing IOT solutions to global stakeholders

Works with Smart City missions and industrial IOT clients across sectors

CHAPTER 2

INTRODUCTION OF IOT AND ARDUINO

2.1 Introduction to IOT (Internet of Things)

The Internet of Things (IOT) refers to a network of physical objects—devices, vehicles, machines, appliances, and more—that are embedded with sensors, software, and other technologies to connect and exchange data with other systems over the internet. These smart devices collect and share real-time data, enabling automation, monitoring, and intelligent decision-making.



Fig 2.1 : IOT Logo

2.2 Key Components of IoT

- Sensors/Devices – Collect data from the environment (e.g., temperature, motion, GPS).
- Connectivity – Data is transmitted through networks such as Wi-Fi, LoRaWAN®, NB-IoT, or Bluetooth.
- Data Processing – The data is analyzed on cloud platforms or edge devices.
- User Interface – Users interact with the system via dashboards, apps, or alerts.

2.3 Importance of IOT in Automation and Real-World Applications

IoT plays a central role in the modern drive toward automation and intelligent systems. It allows machines and systems to perform tasks more efficiently, with real-time monitoring and minimal human intervention. The fusion of IOT with automation is significantly improving performance, safety, and energy efficiency in a wide range of industries.

2.4 Examples of IoT Devices

- Smart thermostats, smart lights, smartwatches
- Surveillance cameras connected to the cloud
- Smart agriculture tools (e.g., soil moisture sensors)
- Industrial machines with real-time monitoring

2.5 Applications of IoT

- **Smart Cities** – Street lighting, traffic control, waste management
- **Agriculture** – Soil monitoring, irrigation control, crop tracking
- **Healthcare** – Patient monitoring, wearable devices
- **Industrial Automation** – Remote monitoring, predictive maintenance
- **Home Automation** – Smart lighting, security systems, energy management

2.6 Advantages of IoT

- Real-time monitoring and control
- Improved efficiency and automation
- Better decision-making with data analytics
- Reduced human effort and operational costs



Fig 2.2 : Applications of IoT

2.7 Introduction to Arduino

Arduino is an open-source electronics platform that makes it easy to build digital devices that interact with the physical world. It is popular in electronics, robotics, IoT, and embedded systems development.

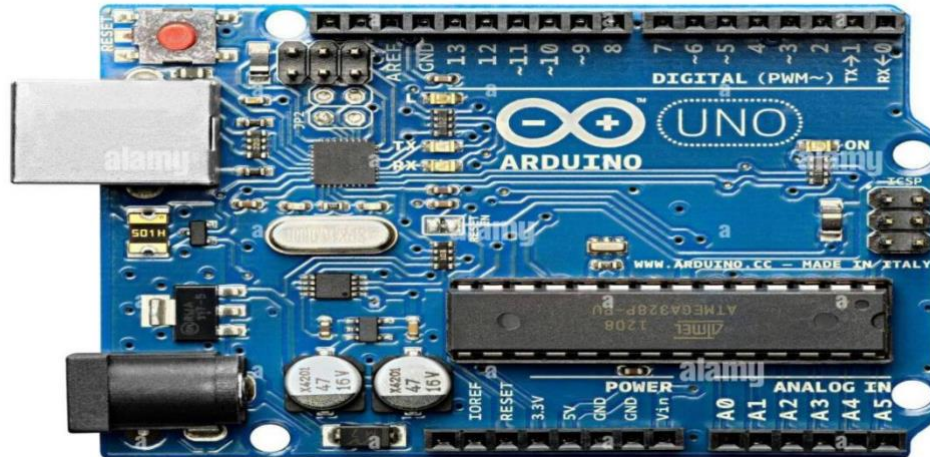


Fig 2.3 : Arduino

2.8 Common Arduino Boards

Board	Features
Arduino Uno	Popular beginner board with ATmega328P MCU
Arduino Nano	Compact version, breadboard friendly
Arduino Mega	More I/O pins and memory
Arduino Leonardo	Built-in USB communication

2.9 Key Features

- Based on microcontrollers (AVR, ARM)
- Uses Arduino IDE for coding (C/C++)
- Connects with hundreds of sensors and modules
- Can interface with other microcontrollers and the internet

2.10 Arduino Components

- Digital and analog pins
- Power supply (5V, 3.3V)
- USB port for programming
- Reset and power indicators

2.11 Applications of Arduino

- Line-following robots
- Smart alarm systems
- Home automation projects
- Motor control and automation

- Educational STEM kits

2.12 Why Integrate Arduino with IoT?

Arduino becomes a smart device when connected to the internet. This allows it to send sensor data to the cloud and receive remote commands — turning it into a complete IoT system.

2.13 COMPARISON TABLE

Feature	IoT	Arduino	Arduino + IoT
Main Role	Data communication and control	Hardware control and sensing	Smart device + cloud connection
Needs Internet?	Yes	No	Yes
Programming	Web/cloud scripting	Arduino C/C++ via Arduino IDE	Arduino + Wi-Fi module + cloud API
Use Case Example	Smart lighting, tracking	Obstacle detection, line-following	Smart irrigation, weather station

CHAPTER 3

BASICS AND FUNDAMENTALS

3.1 Basics of C Programming

C programming is the foundation for most embedded system development. Known for its simplicity, speed, and close-to-hardware capability, C is widely used in writing firmware and microcontroller programs.

3.1.1 Variables

Variables are used to store data in a program. Each variable must be declared with a data type such as int, float, char, etc.

Example:

```
int distance = 10;
```

```
float temperature = 25.6;
```

```
char letter = 'A';
```

3.1.2 Loops

Loops allow repetitive execution of a block of code. Common loops include for, while, and do-while.

Example:

```
for (int i = 0; i < 5; i++) {  
    printf("Count: %d\n", i);  
}
```

3.1.3 Functions

Functions help break a program into smaller, reusable parts. They promote modular programming.

Example:

```
int add(int a, int b) {  
    return a + b;  
}
```

3.1.4 Conditions

Conditional statements control the flow of execution based on logic.

Example:

```
if (distance < 10) {  
    printf("Bin is full\n");  
} else {  
    printf("Bin has space\n");  
}
```

3.1.5 Role in Embedded Systems

In embedded systems like the Smart Bin project, C is used to:

- Interface with sensors (e.g., ultrasonic)
- Control GPIOs (General Purpose Input/Output)
- Implement logic to process sensor data
- Send and receive data over communication protocols like Bluetooth/Wi-Fi
- Optimize code for speed and memory efficiency

Embedded C programs run on microcontrollers (like ESP32), which have limited resources and require efficient code execution.

3.2 Number Systems

A Number System is a way to represent and express numbers using a set of symbols (digits) and rules. It forms the foundation of all computing and digital systems, allowing data to be stored, processed, and transmitted efficiently.

3.2.1 Types of Number Systems**Decimal Number System (Base-10)**

- Uses digits: 0 to 9
- Most commonly used in daily life
- Example: 123, 4567

Binary Number System (Base-2)

- Uses digits: 0 and 1
- Used in computers and digital electronics
- Example: 1011 (binary of 11 in decimal)

Octal Number System (Base-8)

- Uses digits: 0 to 7
- Often used in digital systems as a shorthand for binary
- Example: 17 (octal of 15 in decimal)

Hexadecimal Number System (Base-16)

- Uses digits: 0–9 and letters A–F (A=10, B=11, ..., F=15)
- Widely used in computer programming and memory addressing
- Example: 1F (hex of 31 in decimal)

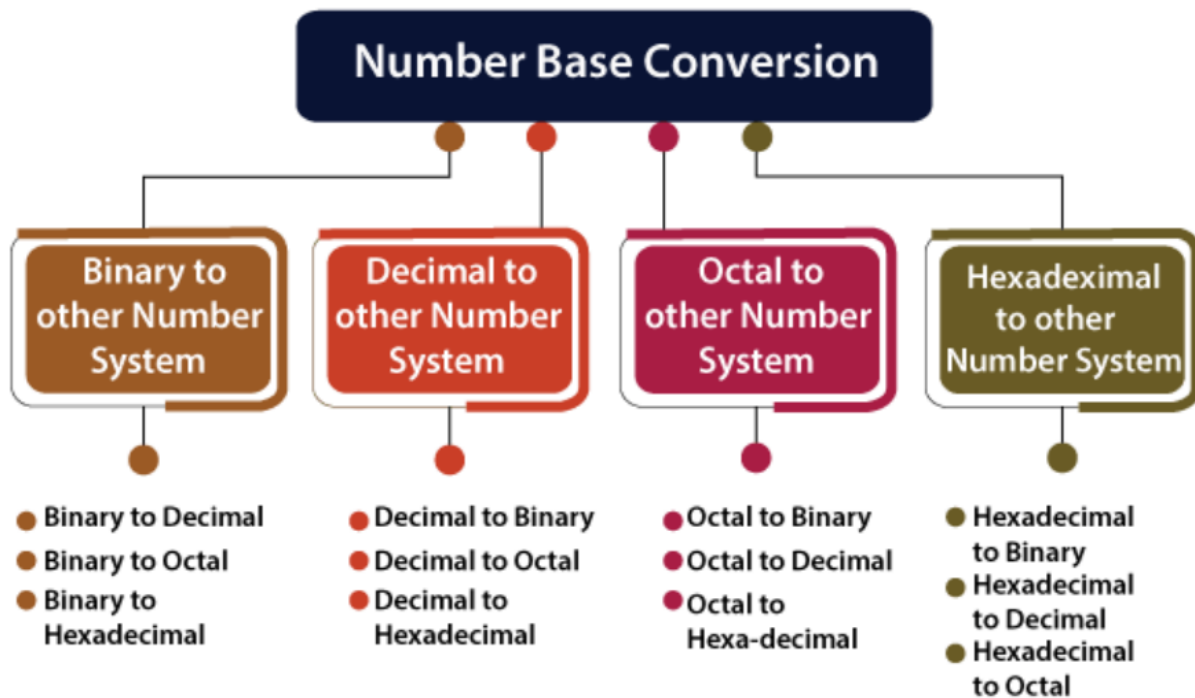


Fig 3.1 : Number Base Conversion

3.3 Applications of Number Systems

Number systems are used extensively in computing, electronics, programming, and digital communication. Each type serves a specific purpose in technology:

1. Binary Number System (Base-2) – 0s and 1s

Used in:

- **Digital electronics and computers** – All data (text, images, videos) is stored in binary.
- **Microcontrollers and processors** – Instructions and data are processed as binary machine code.
- **Logic gates and circuits** – Operate on binary signals (high = 1, low = 0).

2. Decimal Number System (Base-10)

Used in:

- **Human-readable data** – Used in daily life and user interfaces.
- **Input/output operations** – Humans enter and read numbers in decimal form.
- **Calculations in software** – Often converted to/from binary internally.

3. Octal Number System (Base-8)

Used in:

- **Compact binary representation** – Grouping binary digits in sets of 3 (especially in older systems).
- **Digital permissions** – Unix file permission systems use octal values (e.g., 755, 644).
- **Assembly language** – Some low-level programming uses octal for efficiency.

4. Hexadecimal Number System (Base-16)

Used in:

- **Memory addresses and debugging** – Easier to read than long binary strings (e.g., 0x1A3F).
- **Color coding in web design** – HTML/CSS color codes (e.g., #FF5733).
- **Programming and firmware** – Represent instructions and data efficiently.

Decimal System	Binary System	Octal System	Hexadecimal System
0	00000000	000	0
1	00000001	001	1
2	00000010	002	2
3	00000011	003	3
4	00000100	004	4
5	00000101	005	5
6	00000110	006	6
7	00000111	007	7
8	00001000	010	8
9	00001001	011	9
10	00001010	012	A
11	00001011	013	B
12	00001100	014	C
13	00001101	015	D
14	00001110	016	E
15	00001111	017	F

Fig 3.2 : Number Systems

3.4 Logic Gates & Problem Solving

Logic gates are fundamental building blocks of digital circuits. Each gate performs a basic logical function on one or more binary inputs to produce an output.

2.3.1 Basic Logic Gates

Gate	Symbol	Function	Example
AND	$A \cdot B$	Output is 1 only if both A and B are 1	$A = 1, B = 1 \rightarrow \text{Output} = 1$
OR	$A + B$	Output is 1 if either A or B is 1	$A = 0, B = 1 \rightarrow \text{Output} = 1$
NOT	$\neg A$ or A'	Inverts the input	$A = 1 \rightarrow \text{Output} = 0$
NAND	$\neg(A \cdot B)$	Inverse of AND	$A = 1, B = 1 \rightarrow \text{Output} = 0$
NOR	$\neg(A + B)$	Inverse of OR	$A = 0, B = 0 \rightarrow \text{Output} = 1$
XOR	$A \oplus B$	Output is 1 if inputs are different	$A = 1, B = 0 \rightarrow \text{Output} = 1$

3.4.1 Boolean Logic Applications

- Used in **decision-making** circuits and embedded system code
- Create **logic control paths** in programs (e.g., if conditions)
- Design **sensor interfaces** that act only when multiple conditions are met (e.g., if bin is full **AND** lid is closed, then alert)

Example (Boolean logic):

Let A = bin is full

Let B = user nearby

Trigger alert only if $A \text{ AND } B = 1$

3.4.2 Truth Tables

AND Gate Truth Table:

A B Output ($A \cdot B$)

0 0 0

0 1 0

1 0 0

1 1 1

OR Gate Truth Table:

A B Output ($A + B$)

0 0 0

0 1 1

1 0 1

1 1 1

XOR Gate Truth Table:**A B Output ($A \oplus B$)**

0 0 0

0 1 1

1 0 1

1 1 0

3.4.3 Real-World Example – Smart Bin Logic

Let:

- A = bin level > 90%
- B = Bluetooth connection active

System sends alert if both conditions are true:

Output = A AND B

This logic is implemented both in software (if statements) and can be designed in hardware using logic gates.

CHAPTER 4

SENSORS

4.1 Sensors

A sensor is an electronic device that detects and responds to physical inputs from the environment. It converts physical quantities like temperature, light, pressure, motion, or sound into electrical signals that can be measured, analyzed, and used by electronic systems.

4.1.1 Working Principle of a Sensor

Sensors sense a specific input (such as heat or light), convert it into an electrical signal, and send that signal to a microcontroller or processor for interpretation or further action.

4.1.2 Types of Sensors and Their Uses

Sensor Type	Measures	Example Application
Temperature Sensor	Heat or cold (°C/°F)	Thermostats, medical thermometers
IR Sensor	Infrared light (motion, heat)	Obstacle detection, motion sensing
Ultrasonic Sensor	Distance using sound waves	Parking systems, robotic navigation
Light Sensor (LDR)	Light intensity	Street lights, mobile auto-brightness
Gas Sensor	Gas concentration	Gas leak detectors
Touch Sensor	Physical contact	Touchscreens, elevator buttons
Humidity Sensor	Moisture in the air	Weather stations, HVAC systems

4.1.3 Importance of Sensors

- Automation – Sensors are the core of smart devices and systems (IoT, robotics, industry).
- Monitoring – Real-time monitoring of environment, machines, and systems.
- Safety – Help detect hazards (like gas leaks or overheating).
- Convenience – Enable smart features like auto light adjustment, gesture control, etc.

4.2 Temperature Sensor

A temperature sensor is an electronic device that measures the heat or thermal energy of its surroundings and converts it into a readable signal (usually electrical). These sensors are widely used in both industrial and everyday applications to monitor and control temperature.

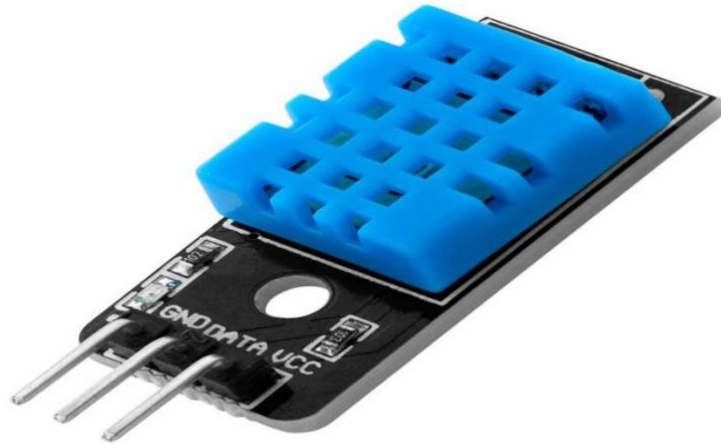


Fig 4.1 : Temperature Sensor

4.2.1 Working Principle

- Temperature sensors work based on physical changes in a material due to temperature:
- Some sensors change their resistance with temperature (e.g., thermistors, RTDs).
- Others produce a voltage or current output (e.g., thermocouples, semiconductor sensors).

4.2.2 Types of Temperature Sensors

Type	Description	Application Example
Thermistor	Resistance changes with temperature	Home appliances, medical devices
RTD (Resistive Temperature Detector)	Highly accurate; used in precision instruments	Industrial process control
Semiconductor Sensor (e.g., LM35, DHT11)	Gives analog or digital output	Arduino projects, weather stations

4.2.3 Applications of Temperature Sensors

- Home Automation – ACs, refrigerators, smart thermostats
- Industrial Automation – Machine temperature monitoring
- Medical Devices – Digital thermometers, incubators
- Automotive Systems – Engine temperature control
- IoT Devices – Remote weather monitoring, smart agriculture

4.3 IR Sensor (Infrared Sensor)

An IR (Infrared) Sensor is an electronic device that detects infrared radiation (IR light) from objects in its surroundings. It is commonly used for object detection, proximity sensing, and motion detection in various electronic and embedded systems.

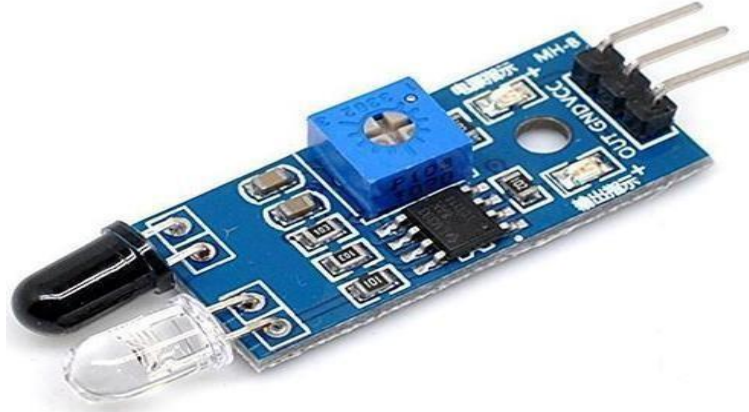


Fig 4.2 : IR Sensor

4.3.1 Working Principle

IR sensors consist of two main components:

IR Emitter (LED) – Emits infrared light.

IR Receiver (Photodiode) – Detects the reflected IR light from nearby objects.

When an object comes near the sensor, the emitted IR light reflects off the object and is received by the detector. Based on the amount of received light, the sensor determines if an object is present.

4.3.2 Types of IR Sensors

Active IR Sensor

- Has both emitter and receiver.
- Used for object detection, distance measurement, line following, etc.

Passive IR Sensor (PIR)

- Detects infrared radiation naturally emitted by warm objects (like the human body).
- Mainly used for motion detection in security systems.

4.3.3 Applications of IR Sensors

- **Obstacle Detection** – In robots and automation systems
- **Line Following Robots** – Detect black/white lines on surfaces
- **Security Systems** – Detect motion using PIR sensors
- **Touchless Controls** – Automatic taps, hand dryers, sanitizer dispensers
- **Remote Controls** – IR used to send signals to TVs, ACs, etc.

4.4 Ultrasonic Sensor

An Ultrasonic Sensor is an electronic device that measures the distance to an object by using ultrasonic sound waves. It is widely used in robotics, automation, and embedded systems for distance measurement, obstacle avoidance, and level detection.



Fig 4.3 : Ultrasonic Sensor

4.4.1 Working Principle

The sensor consists of two main parts:

Transmitter – Emits ultrasonic sound waves (usually at 40 kHz).

Receiver – Detects the echo (reflected sound waves) from the object.

The sensor calculates the time delay between sending and receiving the sound wave and uses the formula:

$$\text{Distance} = \frac{\text{Speed of Sound} \times \text{Time Delay}}{2}$$

(The division by 2 accounts for the sound traveling to the object and back.)

4.4.2 Applications of Ultrasonic Sensors

- **Obstacle Detection** – In robots and autonomous vehicles
- **Distance Measurement** – Parking sensors, object counters
- **Water/Liquid Level Detection** – In tanks or containers
- **Smart Parking Systems** – Detect car presence in a slot
- **Industrial Automation** – Material handling, spacing between products

4.5 Light Sensor (LDR – Light Dependent Resistor)

A Light Sensor, also known as an LDR (Light Dependent Resistor), is a component that detects the intensity of light in the environment. It changes its resistance based on the light falling on it — making it useful for automatic lighting and brightness control systems.

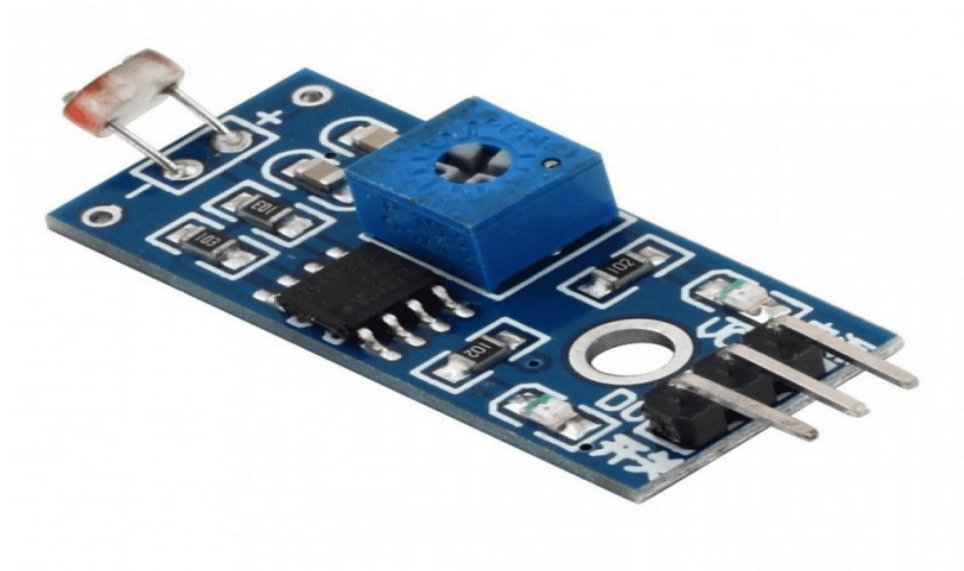


Fig 4.4 : Light Sensor

4.5.1 Working Principle

- An LDR is made of a photosensitive material.
- When light intensity increases, its resistance decreases.
- When light intensity decreases, its resistance increases.
- This change in resistance can be measured and converted into voltage using a circuit (like a voltage divider) to sense light levels.

4.5.2 Applications of Light Sensors (LDRs)

- **Automatic Street Lights** – Turns lights ON in the dark and OFF in daylight
- **Solar Garden Lamps** – Detect darkness to automatically switch ON
- **Brightness Control** – In phones, laptops, and TVs
- **Security Systems** – Detect light changes in a room or area

4.6 Gas Sensor

A Gas Sensor is an electronic device that detects the presence or concentration of gases in the air. It converts the chemical information of a gas (such as carbon monoxide, methane, LPG, or smoke) into an electrical signal, which can be read and analyzed by a microcontroller or monitoring system.



Fig 4.5 : Gas Sensor

4.6.1 Working Principle

Gas sensors typically work by:

- Using a chemical-sensitive material (like tin dioxide or metal oxides).
- When a target gas comes into contact with the material, a chemical reaction occurs that changes its resistance.
- This change in resistance is measured as a voltage change, indicating the gas level.

● Popular Gas Sensors

Sensor Model	Detects	Application Example
MQ-2	LPG, smoke, methane	Gas leak detection systems
MQ-3	Alcohol, benzene, smoke	Breath analyzers, air quality checks
MQ-7	Carbon monoxide (CO)	Indoor air safety
MQ-135	Air quality (CO ₂ , ammonia, etc.)	Pollution monitoring systems

4.6.2 Applications of Gas Sensors

- **Home Safety Systems** – Gas leak detectors in kitchens or factories
- **Industrial Monitoring** – Detect toxic or explosive gases
- **Air Quality Monitoring** – In homes, vehicles, or public spaces
- **Breath Analyzers** – Detect alcohol levels in the breath

4.7 Humidity Sensor

A Humidity Sensor is an electronic device that measures the amount of moisture (water vapor) in the air. It is also called a Hygrometer. These sensors are widely used in weather monitoring, agriculture, HVAC systems, and smart homes.

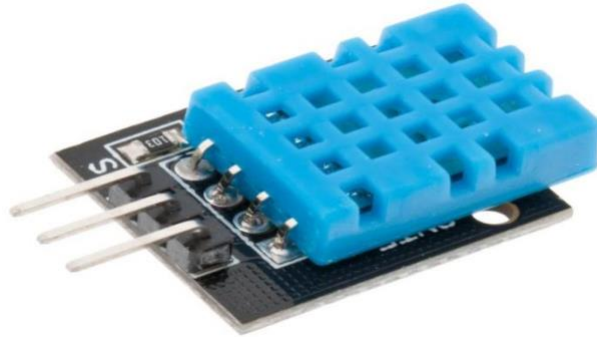


Fig 4.6 : HumiditySensor

4.7.1 Working Principle

Humidity sensors detect humidity using one of the following methods:

- **Capacitive Sensors** – Measure changes in capacitance due to humidity affecting a dielectric material.
- **Resistive Sensors** – Measure changes in resistance when moisture changes conductivity.
- **Thermal Sensors** – Compare temperature differences caused by humid air.

The sensor then provides either an analog or digital output, which can be read by microcontrollers.

4.7.2 Popular Models

Sensor Model	Measures	Output Type	Application Example
DHT11	Humidity + Temp	Digital	Basic weather stations
DHT22 (AM2302)	Humidity + Temp	Digital	Higher accuracy monitoring
SHT31	Humidity + Temp	I2C Digital	Smart home and industrial use

4.7.2 Applications of Humidity Sensors

- **Weather Stations** – Monitor humidity and temperature
- **Agriculture** – Help in irrigation decisions (soil and air humidity)
- **HVAC Systems** – Automatic control of fans, ACs, dehumidifiers
- **Industrial Processes** – Maintain humidity levels for manufacturing

4.8 RFID Sensor (Radio Frequency Identification Sensor)

An RFID Sensor is a device that uses radio waves to identify and track objects wirelessly using an RFID tag. RFID stands for Radio Frequency Identification. These sensors are widely used in security, inventory management, and access control systems.



Fig 4.7 : RFID Sensor

4.8.1 Working Principle

An RFID system consists of two main parts:

- RFID Reader (Sensor Module) – Sends out radio waves and receives data.
- RFID Tag – Contains a microchip with a unique ID and antenna.

When the tag comes close to the reader:

- The reader emits radio waves.
- The tag receives power from these waves (in passive tags).
- The tag sends back its unique ID.
- The reader processes the signal and sends it to a controller or computer.

4.8.2 Types of RFID Tags

Type	Power Source	Range
Passive	No battery	Up to 10 cm – 1 meter
Active	Built-in battery	Up to 100 meters
Semi-passive	Battery-assisted	Intermediate range

4.8.3 Applications of RFID Sensors

- **Access Control** – Door unlocking with ID cards or key fobs
- **Attendance Systems** – Student/employee check-in

CHAPTER 5

PRINTED CIRCUIT BOARD AND THINGZKIT-MINI

5.1 Introduction to PCBs

A **Printed Circuit Board (PCB)** is a board used to support and connect electronic components using conductive tracks, pads, and other features. It forms the backbone of most modern electronic devices.

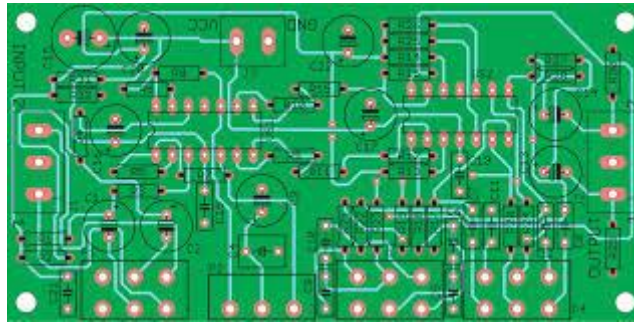


Fig 5.1 : PCB Board

5.2 Importance of PCBs

PCBs are critical in simplifying complex circuits. They reduce the size of electronic systems and increase reliability by minimizing wiring errors. Common applications include:

- Mobile phones
- Computers
- Medical devices
- Industrial equipment
- IoT and embedded systems

5.3 Brief History

- **1903:** First concept of conductive path on an insulated surface
- **1940s:** PCBs used in military radios during WWII
- **Today:** Used in everything from watches to satellites

5.4 Types of PCBs

1. Single-sided PCBs

- Copper tracks only on one side
- Simple and low-cost
- Used in toys, sensors, calculators

2. Double-sided PCBs

- Copper tracks on both sides
- Components can be placed on both sides
- Common in power supplies and instrumentation

3. Multi-layer PCBs

- Multiple layers of copper and insulation
- Very compact and capable of high-density circuits
- Used in laptops, smartphones, and complex communication systems

4. Rigid vs. Flexible PCBs

- Rigid: Solid and inflexible (e.g., computer motherboards)
- Flexible: Can bend and fold (e.g., cameras, wearables)

5. Rigid-Flex PCBs

- Combine both rigid and flexible areas
- Used in aerospace and medical equipment

5.5 PCB Components and Layout

1. Basic Components:

- **Resistors** – limit current
- **Capacitors** – store electrical energy
- **Diodes** – allow current to flow in one direction
- **Transistors** – amplify or switch signals
- **ICs (Integrated Circuits)** – miniaturized circuits in a chip
- **Connectors** – interface with other boards or power supplies

2. PCB Layout Design:

- **Schematic Diagram:** Electronic circuit representation

- **PCB Layout:** The physical placement and routing of components
- **CAD Software:** Tools like Eagle, KiCAD, and Altium are used

3. Design Considerations:

- Signal integrity
- Heat dissipation
- Component spacing
- Ground planes and power routing

5.6 PCB Manufacturing Process

1. Design and Schematic Capture

- Create the circuit diagram
- Translate into a PCB layout using design software

2. Printing the PCB Layout

- Output layout files (Gerber files) for fabrication

3. Etching

- Remove unwanted copper from the board

4. Drilling and Plating

- Drill holes for component leads or vias
- Copper plating for connections

5. Solder Mask and Silkscreen

- Solder mask prevents solder bridges
- Silkscreen adds labels and component info

6. Assembly

- Place and solder electronic components (manual or automated)

7. Testing

- Electrical tests (continuity and isolation)
- Functional testing to ensure proper performance

5.7 Applications:

- **Consumer Electronics:** Phones, TVs, game consoles

- **Automotive:** Engine control units, infotainment
- **Medical:** ECG monitors, pacemakers
- **Industrial:** Robotics, automation systems
- **Aerospace:** Navigation, communication systems

5.8 Advantages of PCBs:

- Compact and lightweight
- Reliable and durable
- Easier maintenance and diagnostics
- Low noise and interference

5.9 Future Trends:

- **Miniaturization** – Smaller, more powerful boards
- **Flexible and wearable PCBs** – Used in smart textiles and health monitoring
- **HDI (High-Density Interconnect)** – Higher performance, smaller footprints
- **Eco-friendly manufacturing** – Lead-free soldering and recyclable materials
- **Integration with AI and IoT** – Smarter PCBs with embedded intelligence

5.10 ThingzKit R Mini Development Board

The **ThingzKit R Mini** is a compact, Arduino-compatible development board designed by **Enthu Technology Solutions India Pvt. Ltd.**, tailored specifically for rapid prototyping and learning in the field of embedded systems and IoT. It combines affordability, simplicity, and powerful interfacing capabilities, making it an ideal choice for students, hobbyists, and early-stage Developer. At its core, the ThingzKit R Mini is based on the ATmega328P microcontroller, the same chip used in the Arduino Uno, providing a familiar platform for developers.

It supports a standard 5V logic level and offers a variety of I/O pins for connecting sensors, actuators, and communication modules. The board is designed to be breadboard-friendly, allowing easy prototyping and circuit assembly without soldering. One of the key highlights of the ThingzKit R Mini is its built-in USB to UART converter, which enables direct programming and serial communication using a standard USB cable, without requiring an external programmer. It also includes onboard voltage regulation circuitry, allowing it to be powered via USB or an external

power source ranging from 6V to 12V. In terms of interfacing, the ThingzKit R Mini provides 14 digital I/O pins, 6 analog input pins, and supports PWM on select digital pins.

This makes it compatible with a wide range of sensors including DHT11, MQ135, IR, Ultrasonic, RFID (MFRC522), OLED displays, and more. I2C and SPI communication protocols are also supported, enabling easy connection with multiple peripheral modules.

The board is widely used in projects like environmental monitoring systems, smart attendance setups using RFID, robotic controls, and sensor data displays. It is particularly valuable in educational labs and hands-on training programs, where learners are introduced to embedded system fundamentals using real-world components.

Overall, the ThingzKit R Mini bridges the gap between basic microcontroller boards and advanced development kits, offering a robust, modular, and cost-effective solution for a wide variety of embedded applications.



Fig 5.2 : ThingzKit R Mini

5.11 Key Features & Components

- **Integrated LED and SHT31 temp/humidity sensor** for environmental measurements.

- **Digital & analog GPIO** pins for flexible sensor/actuator interfacing .
- **Rechargeable battery support**, including:
 1. Protection circuits (over-charge, reverse polarity, short-circuit)
 2. 5 V booster with software enable
 3. Battery-level monitoring and switch for battery vs external power
- **USB (Micro-USB)** port with ESD protection for programming and power

5.12 Wireless Connectivity

- **LoRaWAN** (Class A & C) supporting ABP & OTAA — great for long-range, low-power communication
- **Wi-Fi** (802.11b/g/n, up to 150 Mbps)
- **Bluetooth v4.2** including BR/EDR and BLE

5.13 Why Use ThingzKit R Mini?

- **Portable IoT node:** Battery backup with smart power features.
- **Multi-protocol support:** LoRa, Wi-Fi, Bluetooth all in one compact board.
- **Starter-friendly:** Works with Arduino IDE and includes core sensor/designs.
- **Safety features:** Embedded protections for robust deployment.
- **Locally developed:** Designed and supported in India, with easy access to support.

CHAPTER 6

WIRELESS COMMUNICATION

6.1 Introduction to Wireless Communication

Wireless communication is the transfer of data or power between two or more points that are not connected by an electrical conductor. It enables devices to communicate without needing physical cables, using electromagnetic waves such as radio waves, infrared, and microwaves.

6.1.1 Why Wireless?

- Increases mobility and convenience
- Reduces clutter from wires
- Essential for remote and mobile devices

6.1.2 Brief History:

- 1890s: Wireless telegraphy by Marconi
- 1980s: First generation (1G) cellular networks
- 1990s–2000s: Wi-Fi, Bluetooth, and 3G
- Present: 5G, IoT, and beyond

6.2 Types of Wireless Communication Technologies

Technology	Frequency Band	Range	Usage
Wi-Fi	2.4 GHz / 5 GHz	50-100 m	Internet access
Bluetooth	2.4 GHz	~10 m	Personal devices
Infrared (IR)	< 1 THz	~1 m	TV remotes, simple data transfer
Cellular (3G, 4G, 5G)	Various	Up to km	Mobile phones, IoT
Zigbee	2.4 GHz	10–100 m	Smart homes
LoRa	Sub-GHz (e.g., 868 MHz)	Up to 15 km	Agriculture, smart cities
NFC	13.56 MHz	< 0.1 m	Contactless payments

6.3 Introduction to Wi-Fi

What is Wi-Fi?

Wi-Fi (Wireless Fidelity) is a technology that allows electronic devices to connect to the internet or communicate with each other wirelessly using **radio waves**. It is based on the **IEEE 802.11** standards and is one of the most widely used wireless communication methods in the world.



Fig 6.1 : WiFi Logo

6.3.1 How Wi-Fi Works:

- An internet connection is delivered to a modem, which is connected to a Wi-Fi router.
- The router transmits the internet as a wireless signal.
- Devices like laptops, smartphones, and tablets connect to the signal using their built-in Wi-Fi adapters.

6.3.2 Frequency Bands:

- Wi-Fi typically operates on two main bands:
- 2.4 GHz: Wider coverage, slower speed, more interference
- 5 GHz: Faster speed, shorter range, less interference
- Newer technologies now use 6 GHz (Wi-Fi 6E) for faster and cleaner signals.

6.3.3 Wi-Fi Standards (IEEE 802.11):

Standard	Speed	Frequency	Year
802.11b	11 Mbps	2.4 GHz	1999
802.11g	54 Mbps	2.4 GHz	2003
802.11n (Wi-Fi 4)	600 Mbps	2.4 / 5 GHz	2009

Standard	Speed	Frequency	Year
802.11ac (Wi-Fi 5)	1.3 Gbps+	5 GHz	2013
802.11ax (Wi-Fi 6)	9.6 Gbps	2.4 / 5 / 6 GHz	2019

6.3.4 Wi-Fi Security Protocols:

- **WEP:** Weak, outdated
- **WPA/WPA2:** Common and safer
- **WPA3:** Latest, strongest encryption

6.3.5 Applications of Wi-Fi:

- **Homes:** Internet access for all family devices
- **Offices:** Wireless working environments
- **Schools:** Learning via online platforms
- **Healthcare:** Monitoring and communication tools
- **Public Places:** Hotspots in cafes, airports, malls
- **IoT (Internet of Things):** Smart devices like bulbs, locks, thermostats

6.3.6 Advantages of Wi-Fi:

- **Wireless:** No need for cables
- **Convenient:** Easy to move and use devices
- **Multi-device support:** One network for many gadgets
- **Widely available:** Supported by nearly all modern electronics

6.3.7 Disadvantages of Wi-Fi:

- **Signal interference** from walls, other devices, or networks
- **Security risks** if the network is open or poorly configured
- **Limited range** compared to wired connections
- **Speed drops** when too many devices connect at once

6.3.8 The Future of Wi-Fi:

- **Wi-Fi 6 and 6E** bring faster speeds and better performance in crowded areas
- **Wi-Fi 7 (coming soon)** promises up to 30 Gbps speeds
- **Integration with 5G networks** for seamless wireless connectivity
- **More AI-based network management** to improve performance

6.4 Introduction to Bluetooth

What is Bluetooth?

Bluetooth is a short-range wireless communication technology that allows electronic devices to exchange data over short distances using radio waves. It was originally developed in the 1990s and is now widely used in mobile phones, headphones, computers, and smart devices.

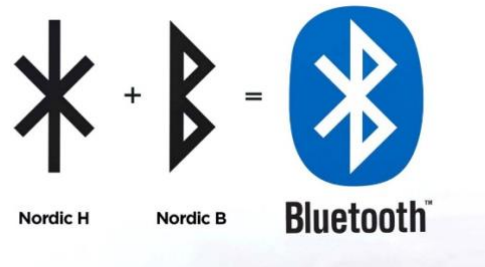


Fig 6.2 : Bluetooth Logo

6.4.1 Key Features:

- Operates in the 2.4 GHz ISM band (Industrial, Scientific, and Medical)
- Low power consumption
- Effective range of about 10 meters (can go up to 100 meters for some devices)
- Supports point-to-point and point-to-multipoint communication

6.4.2 How it Works:

- Bluetooth devices use a master-slave model.
- One device acts as the master, controlling the connection.
- Devices create a piconet, a small network with up to 8 connected devices.

6.4.3 Bluetooth Versions:

Version	Release Year	Key Features
1.0 – 1.2	1999–2003	Basic connectivity
2.0 + EDR	2004	Faster data rates (3 Mbps)
3.0 + HS	2009	High-speed data transfer
4.0	2010	Introduced Bluetooth Low Energy (BLE)
5.0 – 5.3	2016–2021	Longer range, higher speed, better reliability

6.4.4 Main Applications:

- **Wireless audio:** Headphones, earbuds, speakers
- **File transfer:** Between phones, PCs
- **Wearables:** Smartwatches, fitness bands
- **Home automation:** Smart bulbs, door locks
- **Peripheral devices:** Wireless keyboards, mice, printers

6.4.5 Bluetooth Low Energy (BLE):

- Designed for low-power devices
- Used in fitness trackers, IoT sensors, beacons

6.4.6 Security Features:

- Device pairing and authentication
- Encryption to protect data
- Modern Bluetooth uses Secure Simple Pairing (SSP) and AES encryption

6.4.7 Advantages:

- Cable-free communication
- Low power consumption
- Simple pairing process
- Widely supported in consumer devices

6.4.8 Disadvantages:

- Shorter range compared to Wi-Fi
- Lower data transfer speed
- Interference from other 2.4 GHz devices
- Can have security vulnerabilities if not configured properly

6.4.9 Future of Bluetooth:

- Bluetooth 5.4 focuses on power-efficient IoT and industrial applications
- Enhanced audio experiences with LC3 codec in Bluetooth LE Audio
- Mesh networking for smart homes and industry

6.5 Introduction to LoRa

What is LoRa?

LoRa stands for Long Range. It is a wireless communication technology designed for long-distance, low-power, low-data-rate transmission. LoRa is especially suited for Internet of Things (IoT) applications where devices need to send small amounts of data over long distances using minimal power.



Fig 6.3 : LoRa Logo

6.5.1 Key Features:

- Long range: Up to 10–15 kilometers in open areas
- Low power consumption: Ideal for battery-powered devices
- Low data rate: Sends small packets of data
- License-free spectrum: Uses sub-GHz ISM bands (e.g., 868 MHz in Europe, 915 MHz in the USA)

6.5.2 LoRa vs. LoRaWAN:

- **LoRa:** The physical layer (modulation method using Chirp Spread Spectrum)
- **LoRaWAN:** The communication protocol and system architecture that manages how LoRa devices communicate in a network

6.5.3 How LoRa Works:

- A LoRa device (node) collects data (e.g., temperature, motion).
- It sends data wirelessly to a LoRa gateway.
- The gateway forwards the data to a network server (usually via the internet).
- The server processes and routes the data to an application server or database.

6.5.4 LoRa Network Architecture:

- **End Devices (Nodes)** – Sensors or devices that collect and send data
- **Gateways** – Receive wireless LoRa signals and forward them to the server

- **Network Server** – Manages connections, filters data, and handles security

6.5.5 LoRaWAN Classes:

- **Class A:** Most energy-efficient, devices communicate only when needed
- **Class B:** Devices receive scheduled downlink messages
- **Class C:** Devices are always listening (uses more power)

6.5.6 Applications of LoRa:

- **Smart agriculture:** Soil moisture, animal tracking, crop health
- **Smart cities:** Parking sensors, waste bin monitoring, streetlights
- **Industrial IoT:** Asset tracking, energy monitoring
- **Environmental monitoring:** Air quality, flood detection, weather stations
- **Logistics:** Real-time location and condition monitoring

6.5.7 Advantages of LoRa:

- Ultra-long range
- Very low power use
- Excellent for rural or remote areas
- Supports thousands of devices per gateway
- Works indoors and outdoors

6.5.8 Disadvantages of LoRa:

- Low data rates – Not suitable for video or large files
- Latency – Not ideal for time-sensitive tasks
- Requires gateways and network servers
- Possible interference in crowded ISM bands

6.5.9 Future of LoRa:

- Expansion into global smart infrastructure
- Integration with satellite LoRa for global coverage
- Enhanced LoRaWAN security and scalability
- Key enabler in sustainable, energy-efficient IoT systems

6.6 Comparison: LoRa vs. LoRaWAN

Table: Key Differences

Feature	LoRa	LoRaWAN
Definition	A physical layer (modulation technique)	A communication protocol and network architecture
Function	Handles how signals are transmitted wirelessly	Manages how devices communicate within a network
Developer	Semtech Corporation	LoRa Alliance
Technology Layer	Physical Layer (OSI Layer 1)	Network Layer (OSI Layer 2 and above)
Range	Up to 15 km	Inherits range from LoRa
Security	Not handled directly	AES-128 encryption, end-to-end security
Device Classes	Not defined	Class A, B, and C for different power and timing modes
Network Management	Not applicable	Includes gateways, network servers, and application servers
Data Handling	Modulates signal only	Adds routing, session management, security, and application-level support
Usage	Used in many radio-based systems	Specifically designed for IoT networks with LoRa-based communication

6.6.1 Explanation LoRa and LoRaWAN :

- **LoRa** is a modulation technique used to send data wirelessly using chirp spread spectrum. It determines how signals travel between devices and provides long-range communication while using very little power. However, LoRa on its own does not define how devices join networks, route messages, or apply security.
- **LoRaWAN**, on the other hand, is a complete network protocol built on top of LoRa. It defines how devices communicate in a secure and scalable way using gateways, servers, and cloud applications. It includes features like end-to-end encryption, device classes for energy management, and network-level coordination.

CHAPTER 7

MINI PROJECT

7.1 Introduction to RFID Attendance system:

Taking attendance is a basic but important part of managing any classroom or workplace. However, doing it the traditional way — calling out names or using paper registers — can be time-consuming and inefficient. It also opens the door to common issues like errors, buddy punching (marking attendance for someone else), and difficulty keeping long-term records.

This is where **RFID (Radio Frequency Identification)** technology comes in. With an RFID-based attendance system, things become much easier and more accurate. Each person is given a card with a tiny chip (the RFID tag), and when they bring it close to the reader, the system instantly identifies them and records their attendance. It's quick, hands-free, and doesn't require someone to manually enter the data.

What is Arduino and How is it Used?

Arduino is a small electronic board that acts like the "brain" of this system. It can read inputs (like RFID card data), process that information, and take actions like displaying a message, recording data, or even sending it to a computer.

Here's a step-by-step idea of how Arduino is used in this project:

1. **Connecting the RFID Reader:**

The RFID reader (like RC522 or EM-18 module) is connected to the Arduino board. This reader detects the unique ID number stored in each person's RFID card.

2. **Reading the Card:**

When someone places their card near the RFID reader, the reader sends that card's unique ID to the Arduino.

3. **Processing the Data:**

The Arduino is programmed (using the **Arduino IDE**) to recognize this ID. It checks whether the card belongs to a valid person — for example, a student registered in the class.

4. **Taking Action:**

Once verified, Arduino can:

- Mark the person as present by recording the date and time.
- Show their name or a message like "Welcome!" on an LCD screen.
- Send the data to a connected computer or SD card for record-keeping.

- Trigger a buzzer or LED to indicate success or error.

5. Storing or Sending Attendance Data:

The recorded attendance can be stored locally (in Arduino memory or SD card), or transferred to a PC using serial communication. In advanced versions, the data can also be uploaded to cloud platforms or Google Sheets via Wi-Fi modules like **ESP8266**.

The goal of this project is to make attendance tracking easier, faster, and more reliable — especially for places like schools, colleges, or offices where managing attendance manually is a daily challenge. It's a simple idea with big benefits.

7.2 Motivation Behind the Project

An **RFID-based attendance system** makes the whole process quick, automatic, and reliable. Each person just taps their card near the reader, and their presence is recorded instantly — no roll calls, no paperwork. **Our motivation was to create a smart solution that:**

- **Manual attendance takes time** – Calling out names or using registers slows things down, especially in large groups.
- **Prone to errors** – People may forget to mark attendance correctly or make mistakes when entering data.
- **Proxy attendance is common** – Friends marking attendance for each other is a real issue, especially in classrooms.
- **Wastes valuable time** – Teachers or supervisors spend several minutes every day just marking who's present.
- **RFID makes it faster and smarter** – Just tap your card near a reader, and your attendance is recorded instantly.
- **Arduino makes it easy to build** – With Arduino, we can automate the whole system without spending too much money.
- **Better record keeping** – All data is stored digitally, which means it's organized, secure, and easy to access later.
- **Increased security** – Only valid cards work, reducing chances of fake or unauthorized entries.

7.3 Objectives of the RFID attendance system

1. Automate the attendance process

To eliminate manual roll-calling and reduce the time spent on daily attendance.

2. Ensure accurate and reliable records

To avoid human errors and keep clean, organized digital logs of attendance.

3. Prevent proxy attendance

To ensure only the rightful person can mark their attendance using their unique RFID card.

4. Store data digitally

To maintain attendance records in a database or storage device for easy tracking and reporting.

5. Speed up attendance marking

To allow instant check-in by simply tapping an RFID card — faster than traditional methods.

6. Use Arduino for smart control

To use an Arduino microcontroller for reading RFID cards, processing data, and triggering outputs like display or buzzer.

7. Enable real-time monitoring

To allow real-time display or updates of attendance (optional: can be extended to online platforms).

8. Create a low-cost solution

To build the system using affordable components like Arduino, RFID reader, and tags.

9. Easy to scale and upgrade

To design the system in a way that it can be expanded (e.g., to multiple classrooms or departments).

7.4 Advantages of RFID Attendance system:**1. Super Quick**

Just tap your card — that's it. Attendance is marked instantly without wasting time.

2. No Mistakes

Since it's all automatic, there's no risk of someone writing down the wrong name or time.

3. No More Proxy Attendance

Each person has their own unique card, so no one can mark attendance for a friend.

4. Everything is Stored Digitally

The system keeps a record of who came and when — no need to flip through pages or files.

5. Saves Everyone's Time

Teachers or managers don't have to spend minutes or hours on attendance every day.

6. Safe and Secure

The data is stored safely, and there's less chance of it getting lost or changed.

7. Easy to Connect

You can link it to computers, send data to Excel, or even connect it to the internet.

8. Budget-Friendly

Using Arduino and RFID makes it cheap enough for schools, colleges, and small offices.

9. Eco-Friendly

No paper needed — which is good for the planet.

7.5 Components Used

1. thingzkit mini/ESP32 Board

- Main controller to process sensor data and manage communication.

2. Bluetooth Module (Built-in in ESP32)

- Sends bin status and alerts wirelessly to a smartphone or PC.

3. Jumper Wires

- For connecting all components.

7.5.1 thingzkit mini:



Fig 7.2 : thingzkit mini

- **Based on ESP32** – Has built-in Wi-Fi and Bluetooth for wireless projects.
- **USB Powered** – Easy to connect and program using a USB cable.
- **Onboard Sensors** – Includes temperature and humidity sensor (SHT31).
- **LED Indicator** – Built-in LED for simple testing and status display.
- **Multiple GPIO Pins** – Can connect to many sensors and devices.
- **Arduino Compatible** – Works with Arduino IDE; easy to program.

7.5.2 RFID Sensor

1. **What is it?**
A device that uses **radio waves** to read info from RFID tags without touching them.
2. **How does it work?**
It sends a signal to the tag, which replies with stored data like an ID.
3. **What's in it?**
It includes a **sensor (reader)**, an **antenna**, and an **RFID tag**.
4. **Where is it used?**
In **access cards**, **attendance systems**, **libraries**, and **inventory tracking**.
5. **Why use it?**
It's **fast**, **contactless**, and can scan many items at once.
6. **Any drawbacks?**
Can be **costly**, and signals may get blocked by metal or water.
7. **Future potential:**
Will be more powerful with **IoT and smart technologies**.

7.5.3 Bluetooth Module (Built-in in ESP32):

- ESP32 has built-in Bluetooth, so no need for an external module.
- Supports both Bluetooth Classic and Bluetooth Low Energy (BLE).
- Used for wireless communication with phones, laptops, or other devices.
- Allows you to send and receive data (like sensor values or control commands).
- Can connect to Bluetooth apps like Serial Bluetooth Terminal on Android.

7.6 Working Principle of RFID attendance system:

Working Principle of the RFID Attendance System

The RFID Attendance System is designed to automate the process of recording attendance using Radio Frequency Identification (RFID) technology. It provides a fast, secure, and contactless method of identifying individuals and logging their presence in real time.

At the core of the system is the use of RFID cards or tags, each embedded with a unique identification number. These cards are assigned to users (such as students or employees), and when brought near the RFID reader, the system identifies the card and processes the attendance automatically. The entire system is controlled and managed using a microcontroller — typically an **Arduino board**.

Step-by-Step Working Process

1. **Card Scanning**
Each individual carries an RFID card that contains a unique digital ID. When the user

approaches the RFID reader and taps the card near it, the reader emits radio waves and powers the card (if passive), allowing it to transmit its unique ID back to the reader.

2. **Data Reading and Processing**

The RFID reader captures this ID and sends it to the **Arduino microcontroller**, which has been programmed (via Arduino IDE) to handle incoming data. The Arduino then compares the scanned ID with the pre-stored list of authorized IDs.

3. **Attendance Logging**

If the scanned ID matches a registered user, the system marks their attendance. This can also include capturing the **exact time and date**, usually with the help of a **Real-Time Clock (RTC) module**.

4. **User Feedback**

Upon successful verification, the system provides immediate feedback to the user through an **LCD display** (e.g., showing "Attendance Marked") and/or a **buzzer or LED indicator** to confirm a valid scan.

5. **Data Storage or Transmission**

The attendance data is then stored either:

- Locally on an **SD card**, or
- Transmitted to a connected computer, server, or cloud platform (if a **Wi-Fi or Bluetooth module** is used).

7.7 Applications of RFID attendance system

1. Educational Institutions

- Schools, colleges, and universities use RFID systems to track student attendance quickly and accurately.
- Helps teachers save time and reduce paperwork.
- Can also notify parents about student presence via SMS or app integration.

2. Corporate Offices

- Used to monitor employee attendance and working hours.
- Supports payroll systems by automatically calculating working hours.
- Enhances workplace security by allowing only registered staff to access specific areas.

3. Hospitals and Healthcare

- Helps keep track of staff shifts and duty hours.
- Can be used to monitor patient check-ins for appointments or treatments.
- Enhances access control in sensitive areas like ICUs or labs.

4. Industrial and Manufacturing Units

- **Monitors the attendance and shift timing of factory workers.**
- **Increases security by limiting access to authorized personnel only.**

5. Transport and Fleet Management

- **Used for driver attendance tracking and to monitor entry/exit of bus or fleet operators.**
- **Can be integrated with GPS for route tracking and safety logging.**

6. Government and Public Sector Offices

- **Used to ensure punctuality and presence of employees.**
- **Helps maintain transparent attendance records for audits.**

7. Libraries and Labs

- **Tracks entry of students or staff into restricted access areas like libraries, computer labs, or research rooms.**
- **Can also be integrated with systems for book borrowing or lab equipment usage.**

8. Hostels and Dormitories

- **Tracks entry and exit of residents.**
- **Useful for curfew monitoring and ensuring student safety.**

9. Event Management

- **Used to manage entry at seminars, workshops, or conferences.**
- **Helps generate attendance certificates automatically.**

10. Secure Facilities and Research Centers

- **Controls access to sensitive areas by allowing only authorized individuals through RFID-based gates or doors.**
- **Keeps logs of who entered and when.**

7.8 Limitations

1. Limited Reading Range

- **Passive RFID tags have a short detection range, usually just a few centimeters.**

- Users must bring the card very close to the reader, which may cause delays if there's a queue.

2. No Authentication Beyond the Card

- The system only reads the card, not the person.
- If someone else uses your card (card swapping), the system can be tricked unless combined with other verification methods like face recognition or PIN codes.

3. Limited Data Storage

- The RFID tags store only a small amount of data (usually just the unique ID).
- All other information must be stored externally (e.g., on an SD card, computer, or cloud server).

4. Initial Setup and Configuration

- Although Arduino-based systems are affordable, they still require technical knowledge for setup, programming, and troubleshooting.
- Not everyone may be comfortable building or maintaining the system without guidance.

5. No Built-in Internet Connectivity

- Basic RFID systems (with Arduino Uno, for example) do not have Wi-Fi or cloud capabilities unless additional modules (like ESP8266) are added.
- This makes real-time data syncing or remote access more complex.

7.9 Future Scope

- 1. Cloud Storage:** Attendance can be saved online so teachers or managers can check it anytime, anywhere.
- 2. Mobile App:** Students or employees can view their own attendance on a phone app and get alerts if they miss a day.
- 3. Face + RFID:** To prevent misuse, face recognition can be added along with the card — double verification.

4. Smart Reports: The system can create charts and summaries to help understand patterns or frequent absences.

5. Better Security: Encrypted cards and extra verification can make the system more secure.

6. Access Control: RFID can also control doors or gates — only those with attendance marked can enter.

7. Class/Event Link: It can be connected with class schedules or event lists to track who attends what.

7.10 Program:

```
#include <Arduino.h>
#include <ESP8266WiFi.h>
#include <SPI.h>
#include <MFRC522.h>
#include <HTTPSRedirect.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
const char *GScriptId = "ENTER_Deployment ID";
```

```
const char* ssid = "ENTER_WIFI_SSID";
```

```
const char* password = "ENTER_WIFI_PASS";
```

```
const char* host = "script.google.com";
```

```
const int httpsPort = 443;
```

```
String gate_number = "Gate1";
```

```
String url = String("/macros/s/") + GScriptId + "/exec";
```

```
String payload_base = "{\"command\": \"insert_row\", \"sheet_name\": \"Sheet1\", \"values\": \"
```

```
int blocks[] = {4, 5, 6, 8, 9};

#define total_blocks (sizeof(blocks) / sizeof(blocks[0]))

#define RST_PIN 0 // D3
#define SS_PIN 2 // D4
#define BUZZER 4 // D2

MFRC522 mfrc522(SS_PIN, RST_PIN);
MFRC522::MIFARE_Key key;
HTTPSRedirect* client = nullptr;
String student_id;
byte bufferLen = 18, readBlockData[18];

void ReadDataFromBlock(int blockNum, byte* data) {
    for (byte i = 0; i < 6; i++) key.keyByte[i] = 0xFF;

    if (mfrc522.PCD_Authenticate(MFRC522::PICC_CMD_MF_AUTH_KEY_A, blockNum,
    &key, &(mfrc522.uid)) != MFRC522::STATUS_OK) {
        Serial.println("Auth failed");
        return;
    }

    if (mfrc522.MIFARE_Read(blockNum, data, &bufferLen) != MFRC522::STATUS_OK) {
        Serial.println("Read failed");
        return;
    }

    data[16] = data[17] = ' ';
}
```

```

void setup() {
  Serial.begin(9600);
  SPI.begin();
  lcd.init(); lcd.backlight();
  lcd.setCursor(0, 0); lcd.print("Connecting to"); lcd.setCursor(0, 1); lcd.print("WiFi...");
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) delay(1000);
  Serial.println("WiFi Connected");

  lcd.clear();  lcd.setCursor(0, 0);  lcd.print("Connecting  to");  lcd.setCursor(0, 1);
  lcd.print("Google");
  delay(2000);

  client = new HTTPSRedirect(httpsPort);

  client->setInsecure();          client->setPrintResponseBody(true);          client-
>setContentTypeHeader("application/json");

  bool flag = false;
  for (int i = 0; i < 5 && !flag; i++) {
    if (client->connect(host, httpsPort) == 1) {
      lcd.clear(); lcd.setCursor(0, 0); lcd.print("Connected. OK");
      flag = true;
      break;
    }
    delay(1000);
  }

  if (!flag) {

```

```
    lcd.clear(); lcd.setCursor(0, 0); lcd.print("Connection fail");  
    delay(5000);  
    return;  
}  
  
delete client;  
client = nullptr;  
}  
  
void loop() {  
    static bool isClientReady = false;  
    if (!isClientReady) {  
        client = new HTTPSRedirect(httpsPort);  
        client->setInsecure();  
        client->setPrintResponseBody(true);  
        client->setContentTypeHeader("application/json");  
        isClientReady = true;  
    }  
  
    if (!client->connected()) {  
        if (client->connect(host, httpsPort) != 1) {  
            lcd.clear(); lcd.setCursor(0, 0); lcd.print("Disconnected."); lcd.setCursor(0, 1);  
            lcd.print("Retrying...");  
            return;  
        }  
    }  
}
```

```

lcd.clear(); lcd.setCursor(0, 0); lcd.print("Scan your Tag");

mfrc522.PCD_Init();

if (!mfrc522.PICC_IsNewCardPresent() || !mfrc522.PICC_ReadCardSerial()) return;

String values = "", data;
for (byte i = 0; i < total_blocks; i++) {
    ReadDataFromBlock(blocks[i], readBlockData);
    data = String((char*)readBlockData); data.trim();
    if (i == 0) {
        student_id = data;
        values = "\"" + data + ",";
    } else {
        values += data + ",";
    }
}
values += gate_number + "\"}";

String payload = payload_base + values;

lcd.clear(); lcd.setCursor(0, 0); lcd.print("Publishing Data");

lcd.setCursor(0, 1); lcd.print("Please Wait...");

Serial.println(payload);

if (client->POST(url, host, payload)) {
    lcd.clear(); lcd.setCursor(0, 0); lcd.print("Student ID: " + student_id);
    lcd.setCursor(0, 1); lcd.print("Thanks");
} else {
    lcd.clear(); lcd.setCursor(0, 0); lcd.print("Failed."); lcd.setCursor(0, 1); lcd.print("Try Again");
}delay
(5000)}

```

CHAPTER 8

CONCLUSION

This project presents the design and implementation of a **Wi-Fi-based RFID attendance and data logging system** using the **ESP8266 NodeMCU** microcontroller. The system reads user information from RFID cards and uploads this data in real-time to a **Google Sheets spreadsheet** via the internet. It is designed to automate and simplify attendance tracking, access control, or entry management in institutional and secured environments.

The core components of the system include:

- **ESP8266 NodeMCU**: Serves as the main processing and communication unit, handling Wi-Fi connectivity and HTTP requests to Google Sheets.
- **MFRC522 RFID Reader**: Reads user data stored in specific memory blocks of RFID cards.
- **Google Apps Script**: Acts as a lightweight backend API that processes and stores the received data into Google Sheets.
- **LiquidCrystal_I2C 16x2 LCD**: Provides a user interface to display status messages (e.g., “Scan Your Tag”, “Data Published”, or error messages).
- **Buzzer (Optional)**: Can be used for auditory feedback, though not actively used in the code.

The program begins by connecting the ESP8266 to a Wi-Fi network and establishing a secure (though fingerprint validation is bypassed) HTTPS connection to the Google Script Web App. Upon scanning an RFID card, it reads data from designated blocks (such as student ID, name, phone number, etc.), formats this data into a JSON object, and publishes it to Google Sheets using a POST request.

This approach enables real-time data logging without the need for local databases or complex server setups, making the system ideal for educational institutions, office environments, and event management scenarios.

Looking toward the future, there is great potential for **enhancing the system**. The integration of **Wi-Fi or GSM modules** would allow for cloud-based data collection, which can be monitored via a **dedicated mobile app**. This would enable users or authorities to receive push notifications, generate usage reports, and plan better maintenance schedules. Another future enhancement could be the incorporation of **solar panels**, making the system **energy-efficient and environmentally friendly**, especially in outdoor installations.

In conclusion, the Smart Dustbin project proves that **basic IoT components can create**

The developed project successfully demonstrates a fully functional and integrated RFID-based monitoring and data logging system using the ESP8266 NodeMCU microcontroller. The core objective—to create a reliable, real-time system for identifying individuals and recording their details to a cloud-based database—has been met with a robust and scalable solution.

This system utilizes the MFRC522 RFID module to read unique identifiers and stored data from RFID cards, which are commonly used in educational and organizational environments. Each RFID card contains structured data blocks storing personal information such as the student ID, name, contact details, and address. The system is designed to read this data securely and efficiently through block-wise authentication using MIFARE key-based access protocols.

Once the data is retrieved, it is formatted into a JSON payload and transmitted to a Google Sheets document through a secure HTTPS connection using the HTTPSRedirect library. Google Apps Script is used as the backend API, enabling dynamic insertion of data rows into the spreadsheet. This not only facilitates real-time data storage but also ensures easy access, backup, and management of records from any location with internet connectivity.

The ESP8266 microcontroller handles Wi-Fi connectivity, communication logic, and error handling. The system ensures resilience by checking the connection status before each transmission and attempting reconnection in case of failure. This minimizes data loss and enhances operational reliability. Moreover, the use of `setInsecure()` in the HTTPSRedirect library allows for compatibility with evolving SSL certificates, albeit with reduced security due to the absence of certificate fingerprint validation—highlighting a trade-off between ease of development and data security.

The I2C 16x2 LCD provides a user-friendly interface by displaying system status messages such as Wi-Fi connectivity, data publishing status, and acknowledgments. This feedback is particularly helpful for end users during scanning and system interaction, making the system practical for non-technical users.

From a design perspective, the modular and reusable code allows for future extensions—such as integration with databases like Firebase, implementation of multi-gate systems (by assigning unique gate IDs), or adding features like attendance analytics, facial recognition, and mobile notifications.

Overall, the system serves as a cost-effective and scalable solution for automating attendance or access logging tasks in academic institutions, office complexes, and gated facilities. By leveraging open-source hardware and software tools, the project minimizes infrastructure costs while ensuring flexibility, reliability, and remote data access. The outcome not only fulfills the functional and technical requirements but also lays a strong foundation for further IoT-based automation developments.

CHAPTER 9

BIBLIOGRAPHY

1. Enthu Technology Solutions India Pvt. Ltd.
"RFID Based Attendance System – Product Overview and Applications."
Enthu Technology Official Website.
<https://www.enthutech.in>
(Accessed June 2025)
2. Rouse, Margaret.
"Radio Frequency Identification (RFID)."
TechTarget – Internet of Things Agenda,
<https://internetofthingsagenda.techtarget.com/definition/RFID-radio-frequency-identification>
(Accessed June 2025)
3. Finkenzeller, Klaus.
RFID Handbook: Fundamentals and Applications in Contactless Smart Cards and Identification.
3rd Edition, Wiley, 2010.
4. Bhattacharyya, R., Sarma, S. E., & RFID Research Center.
"RFID Technology and its Applications."
MIT Auto-ID Labs White Paper,
Massachusetts Institute of Technology, 2012.
5. Prabhu, Karthikeyan.
"Design and Implementation of RFID Based Attendance System Using Arduino."
International Journal of Advanced Research in Computer Engineering & Technology (IJARCET), Vol. 7, Issue 3, March 2018.
6. Sharma, P., & Singh, R.
"IoT Based Smart Attendance Monitoring System Using RFID."
International Journal of Computer Applications, Vol. 179, No. 7, 2018.